comets impacting from the sky and volcanoes and other tectonic events, helped along by cosmic rays, can congeal liquid hydrocarbons, turning them into some complex organic solid that reflects radio waves back to space. Or maybe something reflective to radio waves is floating on the ocean surface. But liquid hydrocarbons are very under-dense: Every known organic solid, unless extremely frothy, would sink like a stone in the seas of Titan.

Dermott and I now wonder whether, when we imagined continents and oceans on Titan, we were too transfixed by our experience on our own world, too Earth-chauvinist in our thinking. Battered, cratered terrain and abundant impact basins cover other moons in the Saturn system. If we pictured liquid hydrocarbons slowly accumulating on one of those worlds, we would wind up not with global oceans, but with isolated large craters filled, although not to the brim, with liquid hydrocarbons. Many circular seas of petroleum, some over a hundred miles across, would be splattered across the surface—but no perceptible waves would be stimulated by distant Saturn and, it is conventional to think, no ships, no swimmers, no surfers, and no fishing. Tidal friction should, we calculate, be negligible in such a case, and Titan's stretched-out, elliptical orbit would not have become so circular. We can't know for sure until we start getting radar or near-infrared images of the surface. But perhaps this is the resolution of our dilemma: Titan as a world of large circular hydrocarbon lakes, more of them in some longitudes than in others.

Should we expect an icy surface covered with deep tholin sediments, a hydrocarbon ocean with at most a few organic-encrusted islands poking up here and there, a world of crater lakes, or something more subtle that we haven't yet figured out? This isn't just an academic question, because there's a real spacecraft being designed to go to Titan. In a joint NASA/ESA program, a spacecraft called *Cassini* will be launched in October 1997—if all goes well. With two flybys of Venus, one of Earth, and one of Jupiter for gravitational assists, the ship will, after a seven-year voyage, be injected into orbit around Saturn. Each time the spacecraft comes close to Titan, the moon will be examined by an array of